PYTHON

Module 9: Numeric Data

The Three Numeric Types

In Python, numbers are not just digits; they are objects with specific properties. There are three distinct numeric types supported by the language:

• **Int** (Integer): Whole numbers.

• **Float** (Floating Point): Decimal numbers.

• **Complex**: Numbers used for engineering and advanced physics.

Verification: As with all Python objects, you can use the `type() function to confirm which numeric class a variable belongs to.`

Integers (int)

An integer is a whole number, positive or negative, without decimals.

Key Characteristic: In Python 3, integers have **unlimited length**. You can store a number as small as -1 or as large as a googol, limited only by your computer's available memory.

Example:

- small_num = 10
- huge_num = 999888777666555444333

Floating Point Numbers (float)

A float is a number, positive or negative, containing one or more decimals.

Scientific Notation: Floats can also be scientific numbers with an 'e' or 'E' to indicate the power of 10.

Examples:

- Standard: `y = 2.8`
- Positive Power: `x = 35e3  (35,000.0)`
- Negative Power: `z = 12E4  (120,000.0)`

Complex Numbers

Complex numbers are used in specific scientific and mathematical fields. In Python, they are written with a "j" as the imaginary part.

Example:

- `c = 3 + 5j`
- `imaginary_only = 5j`

Note: While rarely used in general web development, they are built-in and ready for data science and engineering tasks.

Numeric Type Conversion

You can convert between types using constructor methods. This is essential when performing math on data from different sources.

Methods:

- float(x) : Converts an integer to a float (e.g., 1 becomes 1.0).
- int(y) : Converts a float to an integer by **truncating** (dropping) the decimal (e.g., 2.8 becomes 2).
- complex(x) : Converts a number to a complex type.

Restriction: You cannot convert complex numbers back into integers or floats.

Generating Random Numbers

Python does not have a `random()` function built directly into the core syntax. Instead, it uses a built-in module named `random`.

To generate a number within a specific range, you must import the module first:

```python
import random
print(random.randrange(1, 10))
```

This will produce a random integer between 1 and 9 (the upper limit is exclusive).

When working with numbers at CodeHive, keep these rules in mind:

- **Precision Matters**: Use floats for currency or measurements, but be aware of floating-point math limitations.

- **Type Safety**: Always check if a variable is an `int` or a `float` before performing operations that require whole numbers (like list indexing).

- **Modular Power**: Explore the `math` module for advanced functions like square roots and trigonometry.